EXPLORATORY ANALYSIS OF MOBILE APP USAGE AND RATINGS USING THE GOOGLE PLAY STORE DATASET

EDA Assignment Report

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1. Abstract

This project explores the Google Play Store dataset to understand mobile application ratings, categories, reviews, installations, prices, and content ratings. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used for data preprocessing, descriptive statistics, group-wise analysis, and visualisation. Charts such as histograms, bar charts, scatter plots, box plots, and a correlation heatmap help identify patterns in the dataset. The reported results include 10,358 app records, an average rating of 4.21, FAMILY as the most common category, Everyone as the most common content rating, and 92.61% free apps.

2. Introduction

Mobile applications are widely used for education, communication, entertainment, shopping, and productivity. Analysing app-store data helps us understand application categories, user ratings, popularity indicators, and pricing patterns. Exploratory Data Analysis (EDA) uses statistical summaries and visualisations to identify patterns and understand a dataset.

3. Problem Statement

The Google Play Store contains applications with different ratings, categories, review counts, installation counts, prices, and content ratings. This project analyses these attributes through data preprocessing and visualisation to identify useful patterns in the app marketplace.

4. Objectives

- Load and inspect the Google Play Store dataset.
- Handle missing values and duplicate records.
- Convert relevant columns into suitable numerical data types.
- Analyse ratings, reviews, installs, prices, and app categories.
- Compare free and paid applications.
- Create visualisations and summarise important findings.

5. Dataset Description

Dataset name: Google Play Store Apps Dataset

CSV file: `googleplaystore.csv`

Source:
https://raw.githubusercontent.com/Sudz24/PlayStore/master/googleplaystore.csv

Important attributes include:

Attribute	Description
App	Application name
Category	Application category
Rating	User rating
Reviews	Number of reviews
Installs	Reported installation count
Price	Application price
Content Rating	Intended audience classification
App Type	Derived category: Free or Paid

6. Data Preprocessing

The following preprocessing steps were performed in the notebook:

1. Loaded the CSV file using Pandas.
2. Inspected the dataset dimensions, columns, and data types.
3. Checked for missing values and handled them using median or mode imputation in the notebook workflow.
4. Removed duplicate rows using `drop_duplicates()`.
5. Converted Reviews into numeric format.
6. Removed commas and plus signs from Installs before numeric conversion.
7. Removed the dollar sign from Price before numeric conversion.
8. Created an App Type column to classify applications as Free or Paid.
9. Generated box plots to inspect potential outliers in Rating, Reviews, Installs, and Price.

7. Exploratory Data Analysis

Descriptive statistics were generated for Rating, Reviews, Installs, and Price. Group-wise analysis compared the average rating, average reviews, and number of apps for free and paid applications. A pivot table compared average ratings by category and app type. A correlation heatmap was used to examine relationships among numerical attributes.

Visualisations

The notebook contains the following visualisations:

1. Histogram of app ratings.
2. Average rating by category.
3. Top 10 categories by number of apps.
4. Price versus rating scatter plot.
5. Free versus paid app count.
6. Top 10 apps by installs.
7. Content rating count plot.
8. Reviews versus rating scatter plot.
9. Average rating by app type.
10. Free versus paid apps across the top 10 categories.
11. Correlation heatmap.

12. Box plots for outlier inspection.

Insert screenshots of the corresponding charts from your Jupyter Notebook in this section.

8. Results and Observations

1. The dataset contained **10,358 app records** after the reported cleaning and duplicate-removal steps.
2. The reported average app rating was **4.21**.
3. **FAMILY** was the most common app category.
4. **Everyone** was the most common content rating.
5. **92.61%** of the analysed apps were free.

9. Business Insights

1. **Free-app dominance:** The high proportion of free apps suggests that developers could investigate advertising or optional premium features as possible monetisation strategies.
2. **Ratings and user experience:** The reported average rating of 4.21 suggests generally positive ratings in the dataset. Developers can examine low-rated apps and user feedback to identify areas for improvement.
3. **Category competition:** FAMILY was the most common category. Developers should compare category-level ratings, app counts, and review volumes before selecting a target market.

10. Recommendations

1. Improve app usability, performance, and reliability through testing, updates, and user-feedback analysis.
2. Evaluate suitable monetisation models, including advertising, paid downloads, or optional premium features.
3. Compare ratings, installs, review volumes, and competition across categories to identify possible market opportunities.

11. Limitations

The findings describe the analysed dataset and may not represent the current Google Play Store. Data quality issues may affect the results. Correlation does not establish causation, and box plots were used for outlier inspection rather than proof that outliers were removed. All figures should be checked against the final notebook before submission.

12. Conclusion

This project used Python-based EDA to analyse Google Play Store application data. The workflow included data inspection, preprocessing, descriptive statistics, group-wise analysis, pivot tables, and visualisation. The reported findings highlight the large share of free apps, the prominence of the FAMILY category, and an average rating of 4.21. The analysis provides a foundation for further investigation into app performance and user experience.

Appendix: Project Files

- Jupyter Notebook: `google playstore.ipynb`
- Cleaned dataset: `googleplaystore_cleaned.csv`
- Report: `Google_Play_Store_EDA_Report.docx` (save this document using the filename above)